Lecture 6 HW

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Original code:

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

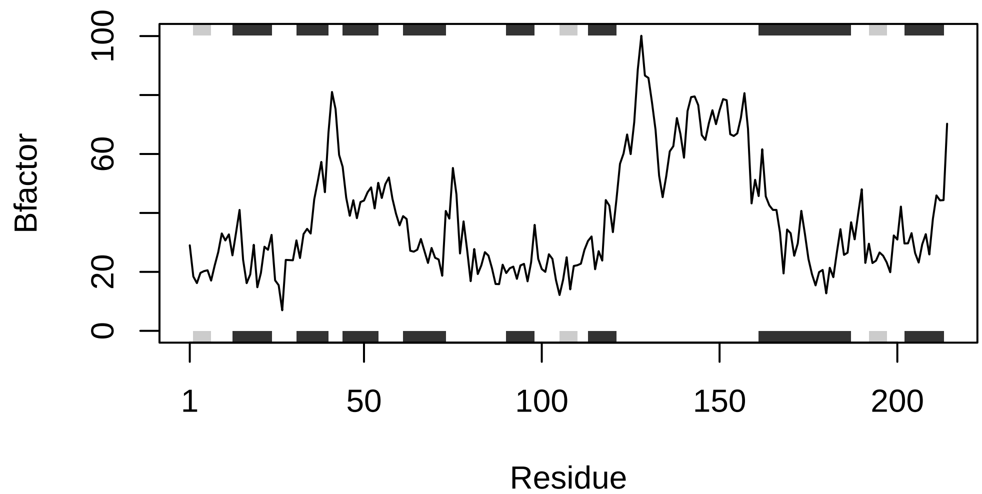
```
s2 <- read.pdb("1AKE") # kinase no drug
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

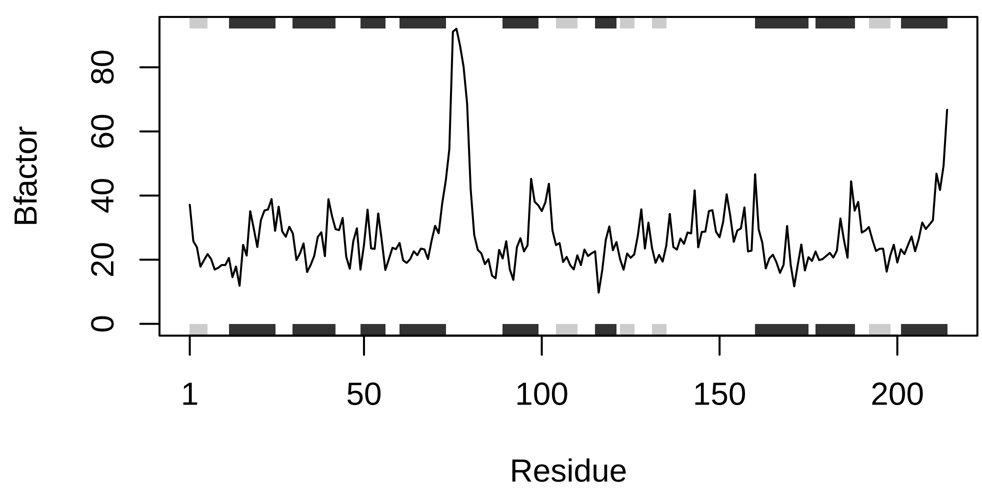
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

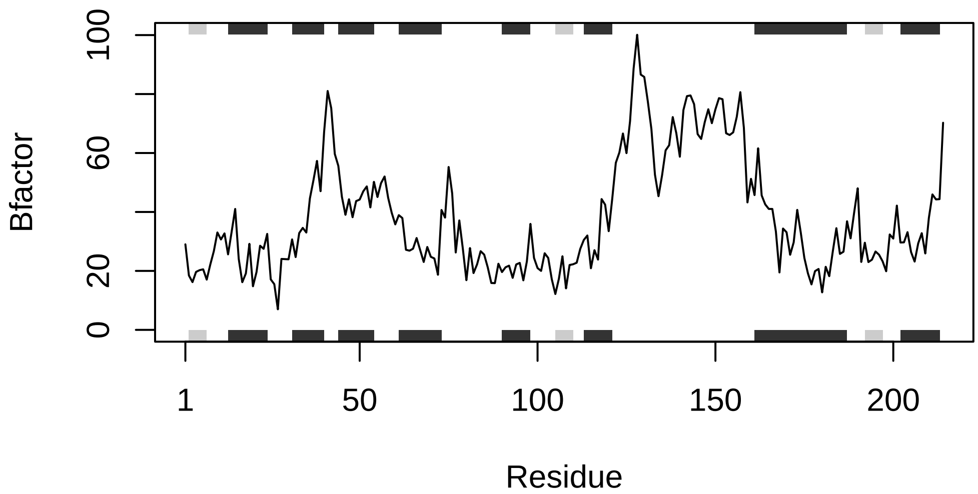
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```




```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```r
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



Q6. How would you generalize the original code above to work with any set of input protein structures?

Code

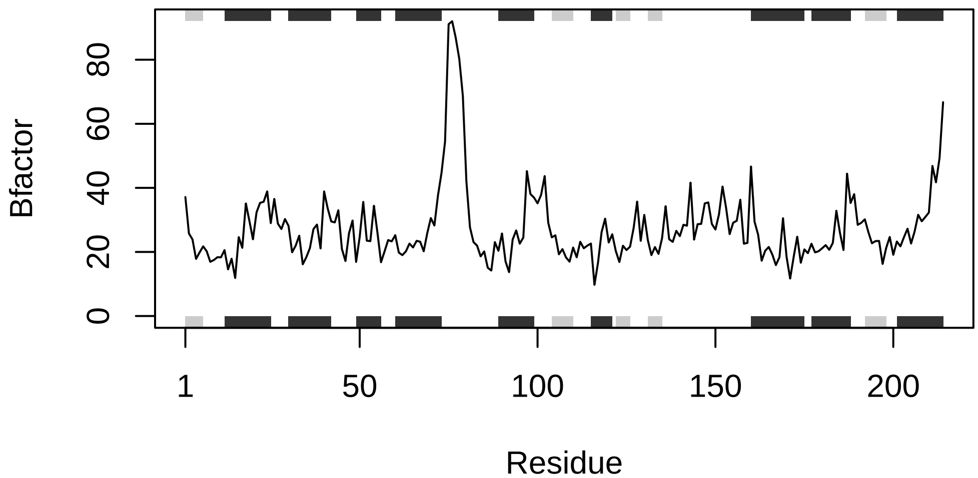
```r
library(bio3d)

process_pdbs <- function(ids, protein_id) {

pdbs <- lapply(ids, read.pdb)

chain <- lapply(pdbs, function(p) trim.pdb(p, chain = "A", elety = "CA"))

bvals <- lapply(chain, function(p) p$atom$b)

i <- match(protein_id, ids)

if(is.na(i)) {
  stop("protein_id not found in ids")
```

```
}

plotb3(bvals[[i]], sse = chain[[i]], typ = "l", ylab = "Bfactor")
}
```

Example output

```
ids <- c("4AKE", "1AKE", "1E4Y")
process_pdbs(ids, "1AKE")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/cn/1kpm2vm94wsc2xd1ym159zch0000gn/T//Rtmp8Wa1TY/4AKE.pdb exists.
Skipping download

  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/cn/1kpm2vm94wsc2xd1ym159zch0000gn/T//Rtmp8Wa1TY/1AKE.pdb exists.
Skipping download

   PDB has ALT records, taking A only, rm.alt=TRUE
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/cn/1kpm2vm94wsc2xd1ym159zch0000gn/T//Rtmp8Wa1TY/1E4Y.pdb exists.
Skipping download
```



Documentation

The important inputs for my function are “ids” and “protein_id”. “ids” is a vector of IDs and “protein_id” is the specific ids to plot. This function reads files, trims the structure to a specific chain and atom, extracts B-factors, and generates a plot for the protein. To use the function, the following code chunk can be utilized:

ids <- c(“4AKE”, “1AKE”, “1E4Y”) process_pdbs(ids, “1AKE”)

The output is a plot for the wanted protein.

Acknowledgement

I used TritonGPT to figure out why my codes don’t work. An example is:

code: library(bio3d) ids <- c(“4AKE”, “1AKE”, “1E4Y”) s <- lapply(ids, read.pdb) chain <- lapply(s, function(p) trim.pdb(p, chain = “A”, elety = “CA”))

bvals <- lapply(chain, function(P) p$atom$b?)

plotb3(bvals[1], sse = chain[1], typ = “l”, ylab = “Bfactor”) Error: unexpected ‘)’ in “bvals <- lapply(chain, function(P) p$atom$b?)”

An example response is:

You're very close. The error is coming from this line:

r

bvals <- lapply(chain, function(P) p$atom$b?) What R is complaining about ? is not valid there, so R stops at the) and says:

unexpected ')'

Two bugs to notice You wrote b? The ? should not be there. Your function argument is P, but inside you use p R is case-sensitive, so P and p are different.